

Section C. Figures of specific lagged effects via DLNM

Load packages

```
packages <- c(
  "readxl", "forecast", "dlnm", "splines", "Metrics", "demography",
  "tseries", "seastests"
)
invisible(lapply(packages, library, character.only = TRUE))
```

Project root

```
find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) {
      return(path)
    }
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()

data_dir <- file.path(base_dir, "Data/Combined_data")
function_dir <- file.path(base_dir, "Code/Function")
```

Load data

```
Y20_64 <- read_xlsx(file.path(data_dir, "Y20_64_combined.xlsx"))
Y65_74 <- read_xlsx(file.path(data_dir, "Y65_74_combined.xlsx"))
Y75_84 <- read_xlsx(file.path(data_dir, "Y75_84_combined.xlsx"))
Y_GE85 <- read_xlsx(file.path(data_dir, "Y_GE85_combined.xlsx"))

age_columns <- c("Y20_64", "Y65_74", "Y75_84", "Y_GE85")
age_titles <- c("Age 20-64", "Age 65-74", "Age 75-84", "Age 85+")

region_specs <- list(
  Athens = list(death_col = 3, exposure_col = 9, source = "Attiki"),
  Lisbon = list(death_col = 4, exposure_col = 10, source = "Lisbon"),
  Rome = list(death_col = 6, exposure_col = 12, source = "Roma")
)
```

Mortality and heat/cold wave inputs

```
build_rate_matrix <- function(spec) {
  out <- cbind(
    Y20_64[[spec$death_col]] / Y20_64[[spec$exposure_col]],
    Y65_74[[spec$death_col]] / Y65_74[[spec$exposure_col]],
    Y75_84[[spec$death_col]] / Y75_84[[spec$exposure_col]],
    Y_GE85[[spec$death_col]] / Y_GE85[[spec$exposure_col]]
  )
  colnames(out) <- age_columns
  out * 52
}

build_wave_data <- function(region, spec) {
  out <- data.frame(
    hot_wave = Y20_64[[paste0(spec$source, "_hot_wave3")]],
    cold_wave = Y20_64[[paste0(spec$source, "_cold_wave3")]]
  )
  colnames(out) <- c(paste0(region, "_hot_wave3"),
    paste0(region, "_cold_wave3"))
  out
}

dat_list <- lapply(region_specs, build_rate_matrix)
wave_list <- Map(build_wave_data, names(region_specs), region_specs)

Athens <- dat_list$Athens
Lisbon <- dat_list$Lisbon
Rome <- dat_list$Rome

Athens_wave <- wave_list$Athens
Lisbon_wave <- wave_list$Lisbon
Rome_wave <- wave_list$Rome
```

Cross-basis matrices

```
source(file.path(function_dir, "crossbasis_mixed_frequency.R"))

utci_list <- list(
  Athens = load_crossbasis_history(base_dir, "Athens", "Attiki"),
  Lisbon = load_crossbasis_history(base_dir, "Lisbon", "Lisbon"),
  Rome = load_crossbasis_history(base_dir, "Rome", "Roma")
)

make_crossbasis <- function(utci) {
  varknots <- equalknots(utci, fun = "ns", df = 3, degree = 3)
  lagknots <- logknots(21, fun = "ns", df = 3)
  crossbasis(
    utci,
    lag = 21,
    argvar = list(fun = "ns", knots = varknots),
    arglag = list(knots = lagknots, df = 3)
  )
}
```

```

}

cb_list <- lapply(utci_list, make_crossbasis)

Athens_cb <- cb_list$Athens
Lisbon_cb <- cb_list$Lisbon
Rome_cb <- cb_list$Rome

```

Load model functions

```

files <- list.files(function_dir, pattern = "\\..R$", full.names = TRUE)
invisible(sapply(files, source))

```

Fit DLNM-LC and DLNM-LL

```

invisible(capture.output({
  DLNM_LC_Athens_fit <- DLNM_LC(
    Athens, Athens_cb, wave_data = Athens_wave,
    tol = 1e-2, max_iter = 20, region = "Athens"
  )
  DLNM_LC_Lisbon_fit <- DLNM_LC(
    Lisbon, Lisbon_cb, wave_data = Lisbon_wave,
    tol = 1e-2, max_iter = 20, region = "Lisbon"
  )
  DLNM_LC_Rome_fit <- DLNM_LC(
    Rome, Rome_cb, wave_data = Rome_wave,
    tol = 1e-2, max_iter = 20, region = "Rome"
  )

  DLNM_LL_fit <- DLNM_LL(
    dat_list = dat_list,
    wave_list = wave_list,
    cb_list = cb_list,
    tol = 1e-2,
    max_iter = 20
  )
}))

```

DLNM residual components

```

lc_temp <- list(
  Athens = as.data.frame(log(as.matrix(Athens)) - DLNM_LC_Athens_fit$final_LC$log_fitted),
  Lisbon = as.data.frame(log(as.matrix(Lisbon)) - DLNM_LC_Lisbon_fit$final_LC$log_fitted),
  Rome = as.data.frame(log(as.matrix(Rome)) - DLNM_LC_Rome_fit$final_LC$log_fitted)
)

ll_temp <- list(
  Athens = as.data.frame(log(as.matrix(Athens)) - DLNM_LL_fit$final_LL$log_fitted$Athens),
  Lisbon = as.data.frame(log(as.matrix(Lisbon)) - DLNM_LL_fit$final_LL$log_fitted$Lisbon),
  Rome = as.data.frame(log(as.matrix(Rome)) - DLNM_LL_fit$final_LL$log_fitted$Rome)
)

```

```

for (region in names(region_specs)) {
  colnames(lc_temp[[region]]) <- age_columns
  colnames(ll_temp[[region]]) <- age_columns
}

```

Fit DLNM components for lagged-effect plots

```

lc_centres <- list(
  Athens = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Lisbon = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Rome = c(Y20_64 = 18, Y65_74 = 20, Y75_84 = 22, Y_GE85 = 22)
)

ll_centres <- list(
  Athens = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Lisbon = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Rome = c(Y20_64 = 18, Y65_74 = 20, Y75_84 = 20, Y_GE85 = 20)
)

fit_dlnm_predictions <- function(temp_data, cb, wave_data, centres, region) {
  hot_col <- paste0(region, "_hot_wave3")
  cold_col <- paste0(region, "_cold_wave3")

  setNames(lapply(age_columns, function(age_col) {
    fit <- glm(
      temp_data[[age_col]] ~ cb + wave_data[[hot_col]] + wave_data[[cold_col]],
      family = gaussian()
    )
    crosspred(
      cb,
      fit,
      cen = centres[[age_col]],
      at = -5:40,
      model.link = "log"
    )
  }), age_columns)
}

lc_pred <- list(
  Athens = fit_dlnm_predictions(lc_temp$Athens, Athens_cb, Athens_wave,
                                lc_centres$Athens, "Athens"),
  Lisbon = fit_dlnm_predictions(lc_temp$Lisbon, Lisbon_cb, Lisbon_wave,
                                lc_centres$Lisbon, "Lisbon"),
  Rome = fit_dlnm_predictions(lc_temp$Rome, Rome_cb, Rome_wave,
                               lc_centres$Rome, "Rome")
)

ll_pred <- list(
  Athens = fit_dlnm_predictions(ll_temp$Athens, Athens_cb, Athens_wave,
                                ll_centres$Athens, "Athens"),
  Lisbon = fit_dlnm_predictions(ll_temp$Lisbon, Lisbon_cb, Lisbon_wave,
                                ll_centres$Lisbon, "Lisbon"),
  Rome = fit_dlnm_predictions(ll_temp$Rome, Rome_cb, Rome_wave,

```

```

    ll_centres$Rome, "Rome")
)

```

Plot helper

```

plot_lag_effects <- function(predictions, ylim) {
  par(
    mfcol = c(2, 4),
    mar = c(5, 5, 2, 1),
    oma = c(1, 1, 3, 1),
    ps = 16,
    cex.lab = 1.2,
    cex.axis = 1.2,
    cex.main = 1.4,
    font.main = 2
  )

  for (age_col in age_columns) {
    plot(
      predictions[[age_col]], "slices",
      var = -5,
      col = 1,
      ylab = "RR",
      ci.arg = list(density = 50, lwd = 2, col = "#AFEEEE"),
      main = paste("UTCI = -5,", age_titles[[which(age_columns == age_col)]]),
      ylim = ylim
    )

    plot(
      predictions[[age_col]], "slices",
      var = 35,
      col = 1,
      ylab = "RR",
      ci.arg = list(density = 50, lwd = 2, col = "#AFEEEE"),
      main = paste("UTCI = 35,", age_titles[[which(age_columns == age_col)]]),
      ylim = ylim
    )
  }
}

```

Figure C.1. DLNM–LC lagged effects for Athens

```

plot_lag_effects(lc_pred$Athens, ylim = c(0.98, 1.05))

```

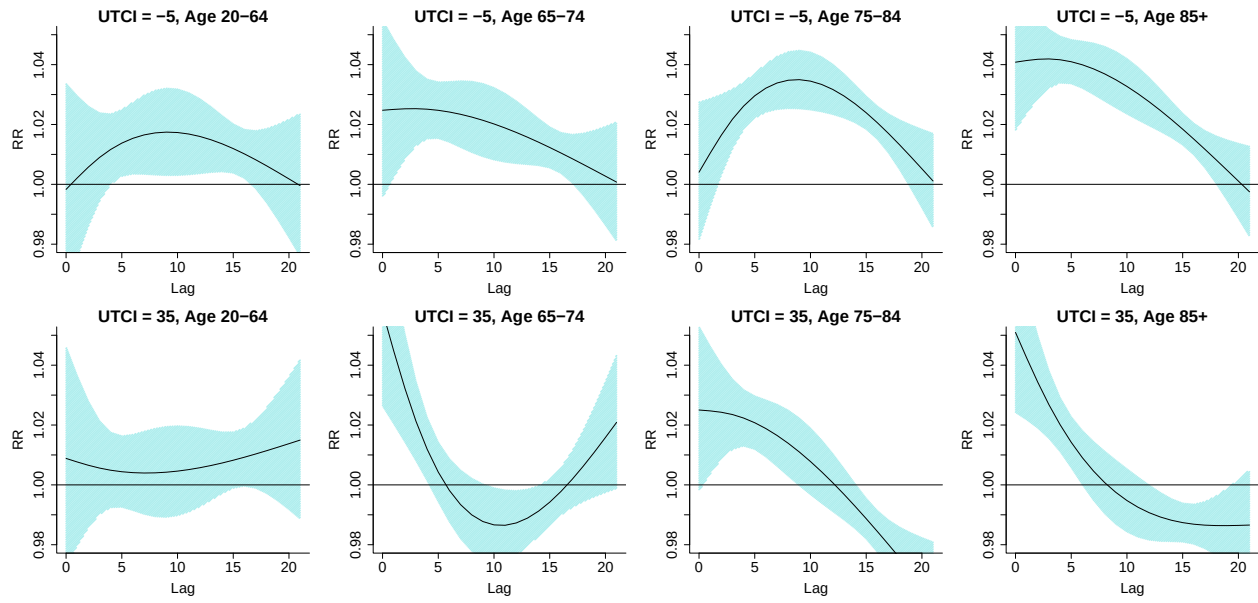


Figure C.2. DLNM-LL lagged effects for Athens

```
plot_lag_effects(ll_pred$Athens, ylim = c(0.98, 1.05))
```

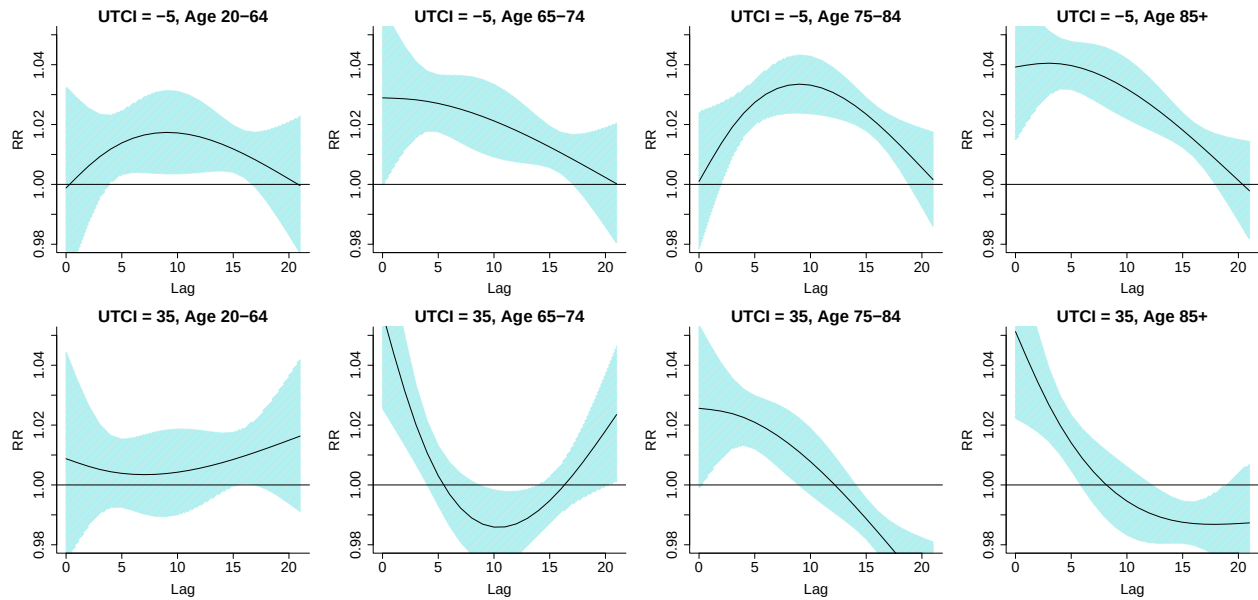


Figure C.3. DLNM-LC lagged effects for Lisbon

```
plot_lag_effects(lc_pred$Lisbon, ylim = c(0.95, 1.2))
```

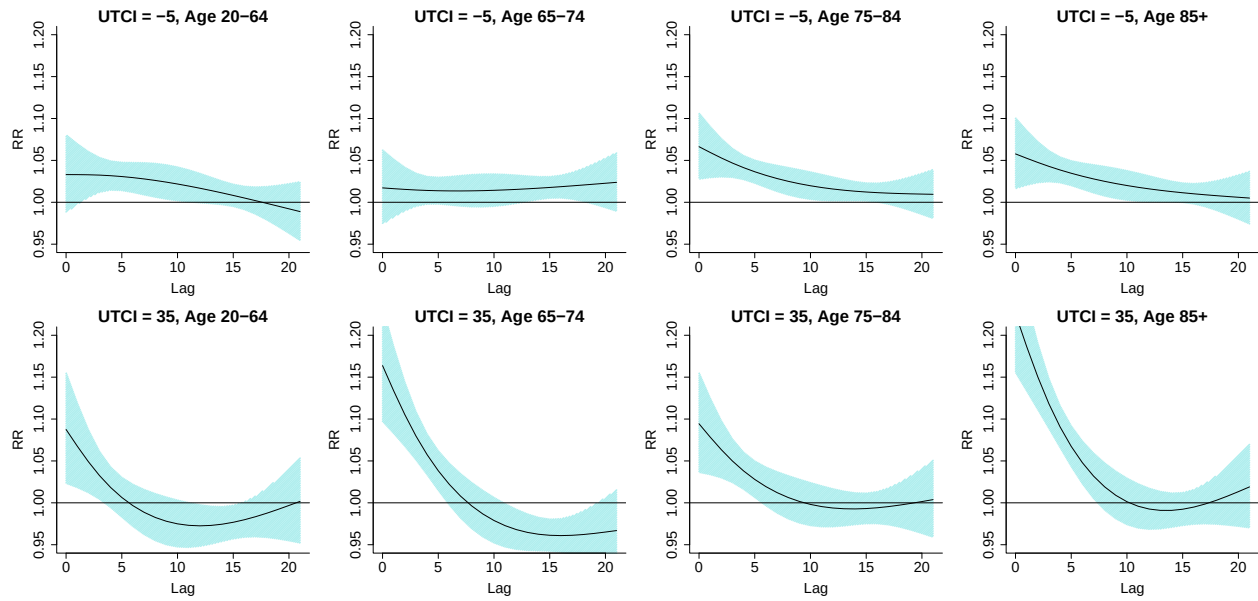


Figure C.4. DLNM-LL lagged effects for Lisbon

```
plot_lag_effects(ll_pred$Lisbon, ylim = c(0.95, 1.2))
```

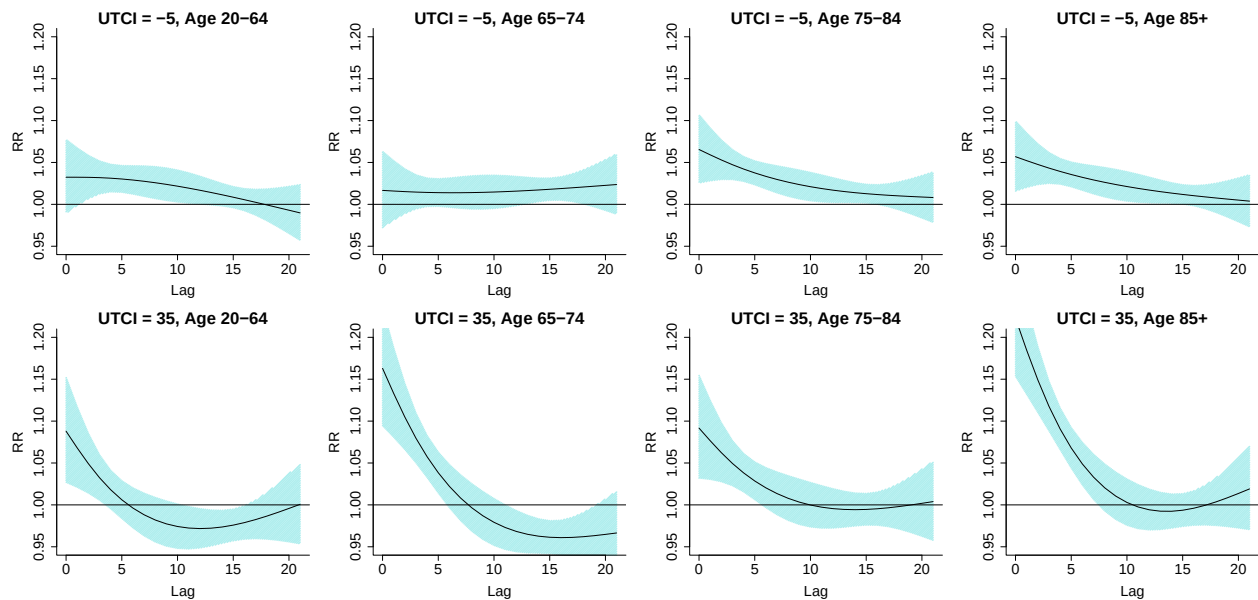


Figure C.5. DLNM-LC lagged effects for Rome

```
plot_lag_effects(lc_pred$Rome, ylim = c(0.95, 1.1))
```

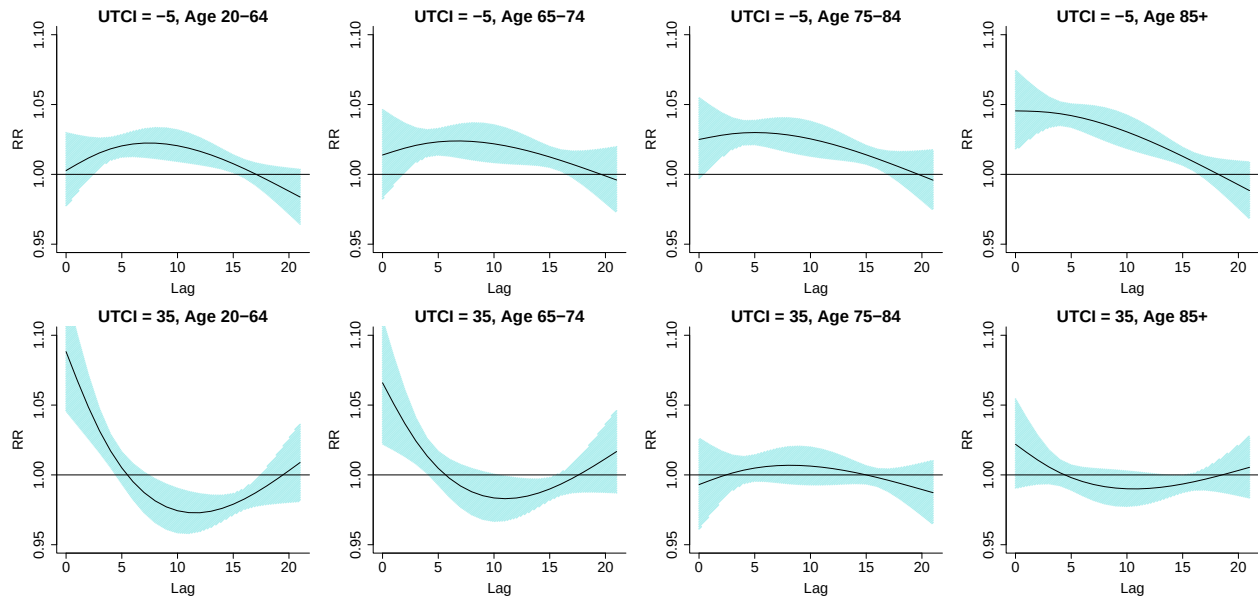


Figure C.6. DLNM-LL lagged effects for Rome

```
plot_lag_effects(ll_pred$Rome, ylim = c(0.95, 1.1))
```

